Moves Before Measure

From a grammar of moves to the geometry of measure

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Abstract

This note separates four notions that the single word *metric* habitually conflates, namely the quadratic form, the inner product, the norm, and the distance function proper, and arranges them not as a list but as a small family of distances in which each step imposes one further constraint, passing from a bare distance to the inner product. It then takes seriously the view that a lattice arrives first with a set of allowable moves and only afterward with a cost assigned to them, so that the geometry native to a lattice is the word metric of its moves, which under a unimodularity condition is precisely a polyhedral norm. From this the round ball is excluded as a native object and admitted only as a limit, so that a lattice can never generate an inner product on its own. The king's walk on $\mathbb{Z}^2$, carrying a single weight on its diagonal step, threads the whole family from ℓ^∞ through a regular octagon to ℓ^1, and at the weight $\sqrt{2}$ the lengths it measures come to live in $\mathbb{Z}[\sqrt{2}]$, whose fundamental unit $1+\sqrt{2}$ and Pell norm return us to the metallic-mean material.

Lattice of Allowable moves & distance family

Picture a chessboard with a king on a1, asked to reach the distant corner h8. The king is a creature of moves rather than of distances, allowed to step to any orthogonally or diagonally adjacent square, and for him the question of how far apart two squares lie has no meaning at all until we decide what a single step is worth.

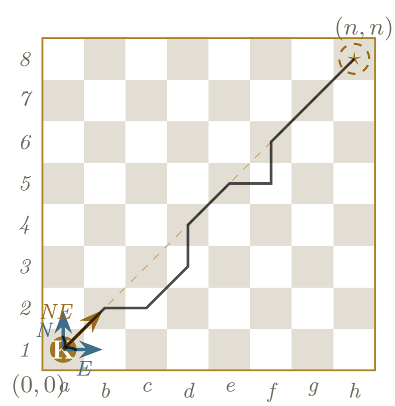

Figure 1: The king walks from a1 to h8 using only East, North, and North-East moves. Three legal first moves are shown in colour, and one completed path is drawn faintly across the board; the dashed line marks the Euclidean geodesic the king can never quite walk.

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If every step is counted equally then the far corner stands eight moves away along any staircase he cares to climb, and the shape of his world is already settled by his repertoire of moves long before anyone produces a ruler. This modest scene holds the whole argument of the present note, that a lattice is given first by its allowable moves and only afterwards by a cost, and that the cost, once chosen, fixes a geometry whose character can be read directly from the moves themselves.

Principle. A lattice possesses no intrinsic notion of length. Its primitive datum is a finite generating set of allowable moves, and distance is a derived concept, obtained only once a cost has been assigned to those generators.

The allowable moves are the short vectors one is permitted to add, and the cost laid over them is the further structure that says how far apart two points stand. The square lattice $\mathbb{Z}^2$ already admits several natural costings that a casual eye runs together: counting unit horizontal and vertical steps gives the Manhattan distance $|a| + |b|$; permitting also the diagonal step and counting moves gives the Chebyshev distance $\max(|a|, |b|)$; insisting that distance be the straight-line length gives the Euclidean $\sqrt{a^2 + b^2}$. These are different answers to a single question, and which of them is appropriate depends entirely on which moves one has declared primary. Reading the move-set as primary, and the cost as a secondary decision laid over it, is the constructive viewpoint of geometric group theory, where a generating set determines a word metric, the least number of generators carrying one point to another.

It helps to see the whole family before defining its members, since the four notions the word *metric* conflates form not a list but a small family of distances, drawn in Figure 2.

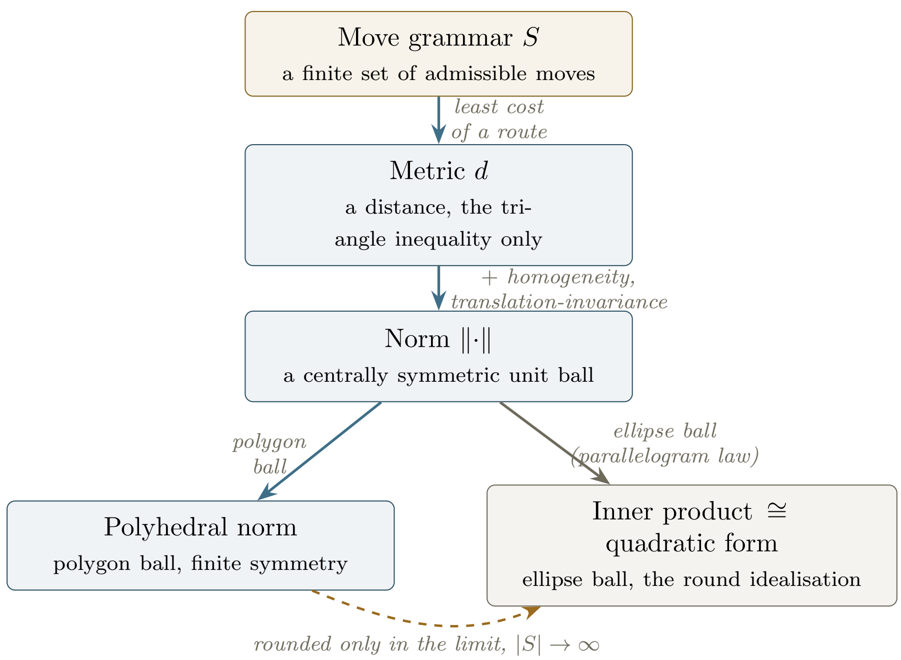

Figure 2: The family of distances generated by a move grammar. Assigning a cost to the moves and taking the least cost of a route gives a metric; because costs scale and add, it is in fact a norm, the gauge of a unit ball. Two distinguished families of norm sit below: the polyhedral norms, with polygon balls of finite symmetry, which are exactly what a finite grammar of moves can reach, and the inner-product norms, with ellipse balls, which are Euclid's.

At its head stands the move grammar, the primitive datum; assigning a cost to the moves and taking the least cost of a route yields a distance, and because those costs scale and add, the distance is already a norm. Each further step downward imposes one more constraint and so names a more special object, until at the foot the family divides into two distinguished kinds, the polyhedral norms that a finite grammar can reach and the inner-product norms, the round balls of Euclid, that it cannot[1]. Reading the figure from the top is therefore a passage from the general to the special, each step adding structure rather than removing it, and the only direction the reader need keep in mind.

With the map in view we may define its nodes precisely.

Definition 1. On a real vector space V: an *inner product* $\langle \cdot, \cdot \rangle$ is a rule combining two vectors into a number that is linear in each argument separately (bilinear), unchanged when the two arguments are swapped (symmetric), and strictly positive on every nonzero vector (positive definite); a *quadratic form* is the map $Q(x) = \langle x, x \rangle$ got by feeding one vector to such a rule twice; a *norm* $\|\cdot\| : V \to \mathbb{R}_{\geq 0}$ is positive away from the origin, absolutely homogeneous in the sense that $\|\lambda x\| = |\lambda| \, \|x\|$, and subadditive in the sense that $\|x + y\| \leq \|x\| + \|y\|$; and a *metric* $d \colon V \times V \to \mathbb{R}_{\geq 0}$ is positive off the diagonal, symmetric, and satisfies the triangle inequality.

The first two nodes are a single datum viewed twice, and it is worth seeing why, since the point is easy to mistake. The quadratic form $Q(x) = \langle x, x \rangle$ records only the inner product of a vector with itself, that is, the squared lengths, and it appears at first to throw away the cross-information $\langle x, y \rangle$ between distinct vectors that carries the angles. Polarisation shows that nothing is in fact lost, because the full pairing can be rebuilt from the squared lengths alone through

$$\langle x, y \rangle = \tfrac{1}{2}\big(Q(x+y) - Q(x) - Q(y)\big),$$

the identity reading $\langle x, y \rangle$ as the amount by which the square of a sum exceeds the sum of the squares. A quadratic form and an inner product are therefore the same object written in two scripts, the one through lengths and the other through pairings, which is why the figure draws them as a single node.

Descending one step, an inner product induces the norm $\|x\| = \sqrt{\langle x, x \rangle}$, and the question of which norms arise this way has a clean answer.

Proposition 2 (Jordan–von Neumann)**.** *A norm $\|\cdot\|$ on a real vector space comes from an inner product if and only if it satisfies the parallelogram law*

$$\|x + y\|^2 + \|x - y\|^2 = 2\,\|x\|^2 + 2\,\|y\|^2\,,$$

in which case the inner product is recovered by $\langle x, y \rangle = \frac{1}{4}\big(\|x+y\|^2 - \|x-y\|^2\big)$. In finite dimension this holds exactly when the unit ball is an ellipsoid.

The parallelogram law is the exact gate to the right-hand branch of the family, since a norm passes through it precisely when its ball is an ellipse and it becomes an inner product. A lattice never reaches that gate on its own, because the distance it builds from a finite set of moves has a polygon for its ball, and a polygon is not an ellipse; the lattice's native geometry is the left-hand branch, the polyhedral norms, and the round ball is approached only as the grammar is enriched without bound. The two branches and the limit that joins them are exactly what Figure 2 records.

This is also where Klein's Erlangen programme enters, and where the present account turns it around. Klein specifies a geometry by a group of admissible transformations and studies the

[1] The two meet only in the limit, as a grammar with more and more moves rounds its polygon toward the circle. The symmetry of the geometry is inherited from the grammar and is therefore finite, which is why the continuously symmetric round ball lies beyond any finite set of moves.

quantities that group leaves invariant. Here the starting point is instead a grammar of moves, and the transformation group emerges from it as the point group permuting those moves, a symmetry derived from the data rather than assumed before it. That emergent group is finite, and a finite symmetry can shape only a polygon, so the continuously symmetric round ball lies beyond any finite grammar; the symmetry of a lattice's geometry, and with it the unreachability of Euclid, are both consequences of the moves rather than premises laid down in advance.

Primitive geometry of moves

We now make the head of the family precise. The word metric is the exact expression of generating distance from moves, the least accumulated cost of a route between two points, and the construction that follows shows it to be, under a mild condition, the polyhedral norm on the left branch of Figure 2.

Definition 3. Let $S \subset \mathbb{Z}^2$ be a finite symmetric set with $S = -S$ and $0 \notin S$, whose elements generate $\mathbb{Z}^2$, and let $c\colon S \to \mathbb{R}_{>0}$ be a cost with $c(s) = c(-s)$. The *weighted word metric* is

$$d_S(0, v) = \min\Big\{ \textstyle\sum_i c(s_i) \ : \ v = \sum_i s_i, \ s_i \in S \Big\},$$

and $d_S(x, y) = d_S(0, y - x)$.

The basic fact is that this combinatorial object is, under a mild hypothesis on how the moves sit in the lattice, nothing other than a polyhedral norm, the gauge of the convex hull of the scaled moves, where the gauge of a convex body is the function returning, for each vector, the factor by which the body must be grown or shrunk so that the vector lands exactly on its boundary, and which for a centrally symmetric body is precisely the norm having that body as its unit ball.

Proposition 4 (word metric as polyhedral gauge). *Write $\widehat{S} = \{ s/c(s) : s \in S \}$ and let $P = \operatorname{conv}(\widehat{S})$, with gauge $g_P(v) = \min\{t \geq 0 : v \in tP\}$. Suppose every point of $\widehat{S}$ lies on the boundary of P, and that any two angularly consecutive elements of $\widehat{S}$ form a basis of $\mathbb{Z}^2$, so that the cones they span are unimodular. Then*

$$d_S(0, v) = g_P(v) \qquad \textit{for all } v \in \mathbb{Z}^2,$$

and the weighted word metric is exactly the polyhedral norm whose unit ball is P.

Sketch. A representation of v by moves of total cost t exhibits v as a sum of t-worth of elements of $\widehat{S}$ after rescaling, so $v \in t\,P$ and $g_P(v) \leq d_S(0, v)$. For the reverse, v lies in some cone spanned by consecutive $\widehat{s}_k = s_k/c(s_k)$ and $\widehat{s}_{k+1} = s_{k+1}/c(s_{k+1})$; unimodularity makes s_k, s_{k+1} a basis, so $v = \alpha s_k + \beta s_{k+1}$ with $\alpha, \beta \in \mathbb{Z}_{\geq 0}$, a path of cost $\alpha c(s_k) + \beta c(s_{k+1})$. Because $\widehat{s}_k, \widehat{s}_{k+1}$ lie on the gauge-one boundary, the edge of P between them lies on the level $g_P = 1$, and a short computation gives $g_P(v) = \alpha c(s_k) + \beta c(s_{k+1})$, whence $d_S(0, v) \leq g_P(v)$. $\square$

Corollary 5. *With unit costs, the rook moves $S = \{\pm e_1, \pm e_2\}$ give the ℓ^1 norm $|a| + |b|$, with P the diamond, and the king moves $S = \{\pm e_1, \pm e_2, \pm(e_1 + e_2), \pm(e_1 - e_2)\}$ give the ℓ^∞ norm* $\max(|a|, |b|)$, *with P the square.*

Remark 6. The unimodular hypothesis is doing real work and cannot be dropped, since for a move-set whose adjacent generators span a cone of determinant larger than one, integer points appear that no minimal continuous path can reach without spending extra moves, and the word metric becomes strictly larger than the gauge of $\operatorname{conv}(\widehat{S})$. For the rook and king sets on $\mathbb{Z}^2$ every adjacent pair has determinant one, which is precisely why the Manhattan and Chebyshev identifications are exact rather than approximate.

Walking like no Ordinary King

Return now to the king of Figure 1 and grant his diagonal step a price of its own. A moment ago every step counted the same; suppose instead that a straight step still costs one while a diagonal step costs w, as though the diagonal were the longer stride it visibly is and ought to be paid for accordingly. The king's sense of how far away things lie bends with that single price, and as we shall see the one number w interpolates the entire family of lattice geometries we have been circling, carrying his world from the square of the plain king-move count out to the diamond of the rook.

Made precise, this is the king's move-set carrying one free parameter, the diagonal cost w against a unit cost for an axis step. Writing $v = (a, b)$ and using the eightfold symmetry to assume $a \geq b \geq 0$, an optimal path for $1 \leq w \leq 2$ clears the smaller coordinate with b diagonal steps and the remainder with axis steps, giving the closed form

$$d_w(0, v) = (a - b) + w\, b = (\max - \min) + w \cdot \min\,.$$

For $1 \leq w \leq 2$ the diagonal is efficient, since one diagonal at cost w does the work of two axis steps at cost 2, and the formula is optimal; at $w = 2$ the diagonal earns nothing and the metric reverts to $\max + \min$, the Manhattan length.

By Proposition 4 the unit ball is $P_w = \operatorname{conv} \widehat{S}$, an octagon with axis vertices $(\pm 1, 0), (0, \pm 1)$ at cost one and diagonal vertices $(\pm \frac{1}{w}, \pm \frac{1}{w})$, the diagonal direction reaching only radius $1/w$ for a unit of cost. The single parameter thus threads a continuous family:

$$w = 1: \text{ the square } \ell^\infty; \qquad w = \sqrt{2}: \text{ a regular octagon}; \qquad w = 2: \text{ the diamond } \ell^1.$$

The middle value is distinguished, since the diagonal vertices $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ then sit at Euclidean radius one alongside the axis vertices, so that all eight vertices lie on the unit circle at equal angular spacing and the octagon is regular. The distinguished weight is the geometric mean $\sqrt{1 \cdot 2} = \sqrt{2}$ of the two degenerate endpoints, and the octagon takes its place as the next polygon in the refining sequence that ran square, hexagon, circle in the companion essays.

Remark 7. A regular octagon carries the dihedral symmetry of order sixteen, while $\mathbb{Z}^2$ admits only the dihedral symmetry of order eight, the crystallographic restriction forbidding any eightfold rotation of a planar lattice, since no fivefold or eightfold turn can carry a lattice to itself. The unit ball is therefore more symmetric than either the lattice or the move-data that produced it, since the axis and diagonal moves remain distinguishable by their differing costs 1 and $\sqrt{2}$, so that the quarter-turn through $45°$ exchanging them is a symmetry of the ball alone. The extra symmetry is a coincidence of the chosen weight rather than a symmetry of the geometry.

Length Arithmetic

At the distinguished weight the values taken by the metric acquire an arithmetic of their own. For $v = (a, b) \in \mathbb{Z}^2$,

$$d_{\sqrt{2}}(0, v) = (\max - \min) + \sqrt{2} \cdot \min \;\in\; \mathbb{Z} + \mathbb{Z}\sqrt{2} = \mathbb{Z}[\sqrt{2}],$$

so that king-path lengths populate the ring $\mathbb{Z}[\sqrt{2}]$, the ring of integers of the real quadratic field $\mathbb{Q}(\sqrt{2})$. This ring brings its own structure, with fundamental unit the silver ratio $1 + \sqrt{2}$ of norm $N(1 + \sqrt{2}) = 1^2 - 2 \cdot 1^2 = -1$, units $\pm(1 + \sqrt{2})^n$, and norm form $N(p + q\sqrt{2}) = p^2 - 2q^2$ of Pell type, which is exactly the metallic-mean and Pell structure met before.

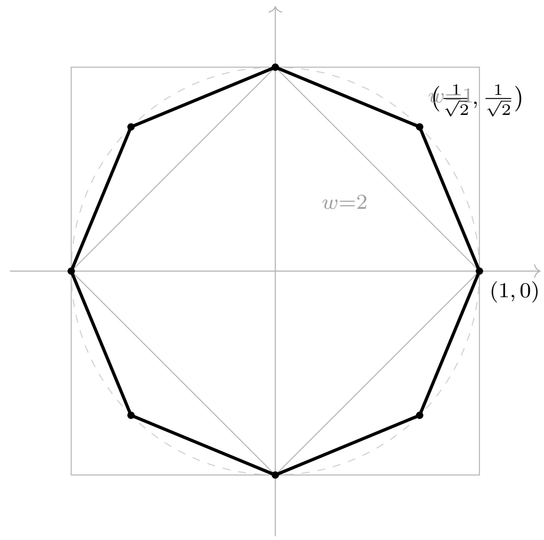

Figure 3: The king's-walk unit balls. The outer square is ℓ^∞ at $w = 1$, the inner diamond is ℓ^1 at $w = 2$, and the heavy octagon is the regular one at $w = \sqrt{2}$, its eight vertices resting on the unit circle. The diagonal vertices slide from the square's corners inward to the diamond's edge midpoints as w runs from 1 to 2.

Remark 8. The word *norm* is now doing two jobs that the reader should hold firmly apart. Through all the geometry above it has meant a length, the degree-one homogeneous gauge $\|\cdot\|$ of a unit ball, which doubles when its argument doubles. In the last display it means instead the field norm $N(p + q\sqrt{2}) = p^2 - 2q^2$, a degree-two multiplicative map carrying an element of $\mathbb{Z}[\sqrt{2}]$ to an ordinary integer and sending a product to a product. The two are genuinely different objects, the first measuring size and scaling linearly, the second a quadratic form of indefinite signature $(1, 1)$ that multiplies rather than measures. What recurs consistently across the two subjects is the quadratic form itself, the geometric $Q(x) = \langle x, x\rangle$ at the structured foot of the family of distances and the arithmetic $p^2 - 2q^2$ governing the units of $\mathbb{Z}[\sqrt{2}]$, one algebraic shape wearing a metric coat in the one setting and a multiplicative coat in the other.

The polyhedral native metric

We are now ready to make the structural claim that motivated our reframing.

Theorem 9. *For any finite symmetric generating set of $\mathbb{Z}^2$ with positive costs, the closed unit ball of the weighted word metric is the convex polygon* $\mathrm{conv}(\widehat{S})$. *The word metric is therefore a polyhedral norm and is never the norm of an inner product, whose unit ball would be an ellipse.*

Proof. The unit ball is $\{v : d_S(0, v) \le 1\}$, which by Proposition 4, where the hypotheses hold, and by convexity of d_S in general, is the gauge-one set of $\mathrm{conv}(\widehat{S})$, a convex polygon with finitely many vertices among the finitely many scaled moves. An ellipse has no vertices and is strictly convex, so no polygon with a finite vertex set is an ellipse, and by Jordan–von Neumann the word metric cannot come from an inner product. $\square$

Observation 10. Euclidean length on a lattice is thus an external refinement rather than a native one, approached as the move-set is enriched without bound, since admitting ever more short vectors with their straight-line costs swells $\mathrm{conv}(\widehat{S})$ toward the disk, the round ball arriving only in the limit and realised by no finite grammar of moves. Read from the side of distance, this is the same demotion that a companion essay performed from the side of area, where the Hodge star was found to supply a canonical measure only for the elliptical ball, so that the square, the hexagon, and now the octagon stand together as the polyhedral geometries a lattice can carry on its own, with the circle reserved for the limit in which the lattice has been forgotten.

Observation 11. The inversion reaches well past the lattice, since any system presented by admissible transitions with accumulated costs carries a distance defined in the same manner, as the least total cost of a route, where costs add along a route and the minimum is taken across routes. This min-plus accumulation is the common spine of Cayley graphs, weighted road networks, the word length of finite-state machines, and the shortest-path and deterministic-control procedures that compute with it, in each of which distance is derived from a grammar of moves rather than posited in advance. Systems that aggregate instead by expectation or by energy, among them Markov chains and resistive networks, replace the minimum with a different combiner and so yield a sibling notion of distance one level removed, a reminder that the move grammar is primitive while the rule for accumulating cost over it is a further choice.

The metric dis-assemblage move

The question arises as to whether our reframing has earned its keep. Is seems so. By making the allowable moves primary and the cost secondary one obtains:

- for the rook, the Manhattan geometry;
- for the king, the Chebyshev geometry;
- and for the Euclideanised king, a one-parameter family of polyhedral norms whose distinguished member is the regular octagon and whose lengths inhabit $\mathbb{Z}[\sqrt{2}]$.

The inner product, the most structured of the norms and the only one whose ball is round, is the single object this construction can never reach, and the four members of the family are best held apart precisely so that one can see why.

The word metric is earlier than the others rather than lesser, the construction from which they are drawn as abstractions, and the king never needed the distinction at all, since he was always going to cross his board in eight honest moves.

Euclidean geometry begins with the ruler and derives its motions, while a lattice begins with its motions and derives whatever ruler we choose to lay against them.